A Gauge Theory of Line Defects in an Elastic Continuum: Full Derivations.

(A Geometrodynamic Model of Particles as Elastic Defects)

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Abstract

This is an extended version of the preprint on a geometrodynamic model of particles as defects of an elastic aether. Unlike the introductory exposition, here *every* structural statement is accompanied by a full derivation: dimensional analysis, intermediate formulas, and, where appropriate, a proof in the form of a theorem. The aim is to make all results reproducible and checkable by the reader "with a pencil". The limits of the model (what is derived, what is assumed, what is open) are traced through each proof, not only in a final section.

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1 Notation and conventions

The three-dimensional elastic continuum (the "aether") is described by a displacement field $u_i(\mathbf{x}, t)$. The dimensional basis is (L, T, M). Greek indices $a, b = 1, 2, 3$ are internal (translational) indices of the gauge group $T(3)$; Latin i, j, k are spatial. Summation over repeated indices is implied.

Definition 1 (Basic elastic quantities)**.** *Shear modulus μ, $[ML^{-1}T^{-2}]$; density ρ, $[ML^{-3}]$; transverse-wave speed $c = \sqrt{\mu/\rho}$, $[LT^{-1}]$; Burgers vector b, $[L]$.*

2 Gauge structure and the Maxwell dictionary

2.1 Distortion field and dislocation tensor

We follow Lazar's notation [3]; all indices are Latin, $i, j, k, l = 1, 2, 3$. Let u_i be the displacement. Incompatibility is introduced by the plastic distortion ϕ_{ij}, the translational gauge field, whose first index is translational. The elastic distortion is

$$\beta_{ij} = \partial_j u_i - \phi_{ij}. \tag{1}$$

The dislocation-density tensor (torsion) is

$$T_{ijk} = \beta_{ik,j} - \beta_{ij,k} = -\left(\phi_{ik,j} - \phi_{ij,k}\right), \tag{2}$$

the second equality holding because $u_{i,kj} = u_{i,jk}$. Thus T is determined by the plastic part alone and is antisymmetric in its last pair of indices, $T_{ijk} = -T_{ikj}$. The Burgers vector of a contour ∂S follows from Stokes' theorem ($\oint A_j\, dx_j = \int_S \epsilon_{klj}\partial_l A_j\, dS_k$):

$$b_i = \oint_{\partial S} \beta_{ij}\, dx_j = \int_S \alpha_{ik}\, dS_k, \qquad \alpha_{ik} \equiv \epsilon_{klj}\beta_{ij,l} = \tfrac{1}{2}\epsilon_{klj}T_{ilj}. \tag{3}$$

Here α_{ik} is Nye's tensor. The last equality holds because only the part of $\epsilon_{klj}\beta_{ij,l}$ antisymmetric in (l, j) survives, and $\frac{1}{2}(\beta_{ij,l} - \beta_{il,j}) = \frac{1}{2}T_{ilj}$. The sign of b is a convention; the classical literature [9] often uses the opposite one.

Lemma 1 (Dimension of the magnetic charge)**.** *In the elastic basis the magnetic charge, defined as the flux of the dislocation density $q_m = \int_S \alpha_{ik}\, dS_k$ (3), has the dimension of length.*

Proof. The distortion β_{ij} is dimensionless (a displacement gradient minus the dimensionless ϕ). From (2), $[T_{ijk}] = [\alpha_{ik}] = L^{-1}$, so $[q_m] = L^{-1} \cdot L^2 = L$. □ □

Remark 1. *Lemma 1 removes the apparent conflict with SI, where magnetic charge is measured in $A \cdot m$: in a consistent elastic theory the ampere is not a primary unit, and the dimension of the charge is dictated by the theory itself.*

2.2 Dislocation current

Let v_i be the physical (in general incompatible) velocity of the medium. The dislocation current is [3, eq. (1)]

$$I_{ij} = -v_{i,j} + \dot{\beta}_{ij}, \qquad [I_{ij}] = T^{-1}. \tag{4}$$

The density T and the current I are kinematically related by the Bianchi identities derived below, (11). In form these are the homogeneous Maxwell equations, whence the dictionary $\mathbf{B} \leftrightarrow T$ (more precisely α), $\mathbf{E} \leftrightarrow I$ (Table 1).

2.3 Full derivation of the quasi-Maxwell system

We give the derivation in detail, following Lazar's linear isotropic theory in index notation [3] (the differential-form version and the nonlinear case are in [2]). In this subsection all indices are Latin and spatial, $i, j, k, l = 1, 2, 3$; summation over repeated indices is implied.

Kinematics. The basic fields are the elastic distortion β_{ij} (dimensionless) and the physical velocity v_i $[LT^{-1}]$; both are incompatible. The dislocation density and the dislocation current are [3, eq. (1)]

$$T_{ijk} = \beta_{ik,j} - \beta_{ij,k}, \qquad I_{ij} = -v_{i,j} + \dot{\beta}_{ij}, \tag{5}$$

with $\beta_{ij,k} \equiv \partial_k \beta_{ij}$, $\dot{\beta}_{ij} \equiv \partial_t \beta_{ij}$; $[T_{ijk}] = L^{-1}$, $[I_{ij}] = T^{-1}$. These are the same definitions as (2) and (4); the first index is translational.

Lagrangian. The Lagrangian density has the bilinear form [3, eq. (3)]

$$\mathcal{L} = T - W = \tfrac{1}{2}\, p_i v_i + \tfrac{1}{2}\, D_{ij} I_{ij} - \tfrac{1}{2}\, \sigma_{ij} \beta_{ij} - \tfrac{1}{4}\, H_{ijk} T_{ijk}. \tag{6}$$

Scalar character. All indices are contracted in each term: one in the first, two in the second and third, three in the fourth. No index is free, and $\mathcal{L}$ is a scalar. Each term has the dimension of an energy density $[ML^{-1}T^{-2}]$: $[p_i v_i] = \frac{M}{L^2 T} \cdot \frac{L}{T}$, $[D_{ij} I_{ij}] = \frac{M}{LT} \cdot \frac{1}{T}$, $[\sigma_{ij}\beta_{ij}] = \frac{M}{LT^2}$, $[H_{ijk} T_{ijk}] = \frac{M}{T^2} \cdot \frac{1}{L}$.

Responses. The canonically conjugate quantities are [3, eq. (4)]

$$p_i = \frac{\partial \mathcal{L}}{\partial v_i}, \quad \sigma_{ij} = -\frac{\partial \mathcal{L}}{\partial \beta_{ij}}, \quad D_{ij} = \frac{\partial \mathcal{L}}{\partial I_{ij}}, \quad H_{ijk} = -2\frac{\partial \mathcal{L}}{\partial T_{ijk}} : \tag{7}$$

the momentum, the force stress, the dislocation momentum flux and the pseudomoment stress. The factor 2 in H compensates the coefficient $\frac{1}{4}$ and the antisymmetry $T_{ijk} = -T_{ikj}$.

Constitutive relations. For a linear isotropic medium [3, eqs. (10)–(13)]

$$\begin{aligned} p_i &= \rho\, v_i, \\ \sigma_{ij} &= \lambda\, \delta_{ij} \beta_{kk} + \mu(\beta_{ij} + \beta_{ji}) + \gamma(\beta_{ij} - \beta_{ji}), \\ D_{ij} &= d_1\, \delta_{ij} I_{kk} + d_2(I_{ij} + I_{ji}) + d_3(I_{ij} - I_{ji}), \\ H_{ijk} &= c_1\, T_{ijk} + c_2(T_{jki} - T_{kji}) + c_3(\delta_{ij} T_{llk} - \delta_{ik} T_{llj}). \end{aligned} \tag{8}$$

These are ten constants: ρ and nine moduli, $[\lambda, \mu, \gamma] = ML^{-1}T^{-2}$, $[d_{1,2,3}] = ML^{-1}$, $[c_{1,2,3}] = MLT^{-2}$. Non-negativity of the energy $T + W \geq 0$ requires [3, eq. (14)]: $\rho, \mu, \gamma \geq 0$, $3\lambda + 2\mu \geq 0$; $d_2, d_3 \geq 0$, $3d_1 + 2d_2 \geq 0$; $c_1 - c_2 \geq 0$, $c_1 + 2c_2 \geq 0$, $c_1 - c_2 + 2c_3 \geq 0$.

Relation to the elasticity tensor. The elastic part is $W_{\mathrm{el}} = \frac{1}{2}\sigma_{ij}\beta_{ij} = \frac{1}{2} C_{ijkl} \beta_{ij} \beta_{kl}$ with

$$C_{ijkl} = \lambda\, \delta_{ij}\delta_{kl} + (\mu + \gamma)\, \delta_{ik}\delta_{jl} + (\mu - \gamma)\, \delta_{il}\delta_{jk}.$$

Each of the four indices of C is contracted with an index of one of the two distortions. For $\gamma = 0$ this is the usual isotropic Lamé tensor. The constant γ assigns energy to the antisymmetric (rotational) part $\beta_{[ij]}$; it is absent in classical elasticity, where the distortion is a displacement gradient and a rotation carries no energy.

Remark on the earlier form. In the previous version of this paper the elastic term was written as $\frac{1}{2}C_{ijkl}\beta^i{}_a\beta^{ja}$. There the indices k, l are not contracted, so the expression is not a scalar; moreover, $D_{ab}I^a_i I^{bi}$ and $A_{ab}T^a_{ij}T^{bij}$ had one constant each instead of three. The present form (6)–(8) corrects both points. The earlier simplified form is the special case $d_1 = 0$, $d_2 = d_3$ and $c_2 = c_3 = 0$. Since the results of the following sections use only the structure of the equations and not the specific values of d_i, c_i, they are not affected by this correction.

Equations of motion. The Euler–Lagrange equations with respect to v_i and β_{ij} [3, eqs. (5)–(6)], with external sources p^0_i, σ^0_{ij} (null Lagrangian $\sigma^0_{ij}\beta_{ij} - p^0_i v_i$, with $\dot{p}^0_i - \sigma^0_{ij,j} = 0$), take in terms of the responses (7) the form [3, eqs. (7)–(8)]

$$D_{ij,j} + p_i = p^0_i, \qquad \text{(dislocation momentum balance; Gauss-law analogue)} \tag{9}$$

$$\dot{D}_{ij} + H_{ijk,k} + \sigma_{ij} = \sigma^0_{ij}. \qquad \text{(stress balance; Ampère–Maxwell analogue)} \tag{10}$$

Differentiating (10) with respect to x_j and (9) with respect to t and subtracting gives the integrability condition $\dot{p}_i - \sigma_{ij,j} = 0$, the ordinary equation of motion of the elastic medium [3, eq. (9)]. The terms $D_{ij,j}$ and $H_{ijk,k}$ cancel by symmetry.

How this pair corresponds to Maxwell's equations. In (10) the term $\dot{D}$ corresponds to the displacement current $-\frac{1}{c}\dot{\mathbf{D}}$, and $H_{ijk,k}$ to the curl $\nabla \times \mathbf{H}$ (the divergence of a tensor antisymmetric in jk is equivalent to the curl of the vector $\frac{1}{2}\epsilon_{jkl}H_{ijk}$). The stress σ_{ij} plays the role of the current, and p_i in (9) the role of the charge density. Thus the sources of the dislocation field are the momentum and the stresses of the elastic medium.

Bianchi identities. The definitions (5) imply identically [3, eq. (2)]

$$\epsilon_{jkl}\, T_{ijk,l} = 0, \qquad \dot{T}_{ijk} + I_{ij,k} - I_{ik,j} = 0. \tag{11}$$

The first is the analogue of $\nabla \cdot \mathbf{B} = 0$ (no sources of dislocation density: a dislocation line cannot end inside the medium). The second is the analogue of Faraday's law. Check of the second: $\dot{T}_{ijk} = \dot{\beta}_{ik,j} - \dot{\beta}_{ij,k}$ and $I_{ij,k} - I_{ik,j} = -v_{i,jk} + \dot{\beta}_{ij,k} + v_{i,kj} - \dot{\beta}_{ik,j}$; the sum vanishes because $v_{i,jk} = v_{i,kj}$. Comparing (8)–(11) with Maxwell's equations gives the dictionary of Table 1 (below): the strengths T, I correspond to $\mathbf{B}, \mathbf{E}$, and the responses H, D to $\mathbf{H}, \mathbf{D}$.

Peach–Koehler force. In index notation the force density acting on the dislocations is

$$f_k = \sigma_{ij}T_{ijk} - p_i I_{ik}, \tag{12}$$

which is the component form of eq. (31) of [2], $f^a = (e_a \lrcorner I^b)p_b + (e_a \lrcorner T^b) \wedge \Sigma_b$. The derivation of (12) is ours and is not taken from [3]. Substituting (2) and (4):

$$f_k = \sigma_{ij}\beta_{ik,j} - p_i\dot{\beta}_{ik} - \sigma_{ij}\beta_{ij,k} + p_i v_{i,k}.$$

Using the elastic equation of motion $\dot{p}_i = \sigma_{ij,j}$ (the integrability condition of (9)–(10)), the first two terms combine: $\sigma_{ij}\beta_{ik,j} - p_i\dot{\beta}_{ik} = \partial_j(\sigma_{ij}\beta_{ik}) - \partial_t(p_i\beta_{ik}) - \beta_{ik}(\sigma_{ij,j} - \dot{p}_i) = \partial_j(\sigma_{ij}\beta_{ik}) - \partial_t(p_i\beta_{ik})$. For a homogeneous linear medium $C_{ijlm} = C_{lmij}$, so $\sigma_{ij}\beta_{ij,k} = \partial_k W$, and $p_i v_{i,k} = \rho v_i v_{i,k} = \partial_k T$. Hence

$$f_k = -\partial_t(p_i\beta_{ik}) + \partial_j(\sigma_{ij}\beta_{ik} - \delta_{jk}(W - T))\,. \tag{13}$$

The force density is the divergence of a four-dimensional Eshelby-type tensor. Therefore f_k is the configurational force on the defects, and $\int f_k\, dV$ over a region containing a defect reduces to a flux through its boundary. Here $W - T = -\mathcal{L}$ appears, i.e. (13) is the Eshelby tensor $\mathcal{L}\delta_{jk} + \sigma_{ij}\beta_{ik}$ of dynamic elasticity. The minus sign in front of $p_i I_{ik}$ in (12) is fixed precisely by requiring this identity. We checked it numerically on random smooth fields: with this sign the residual $-\beta_{ik}(\sigma_{ij,j} - \dot{p}_i)$ vanishes on the equations of motion, and with a plus sign it does not. The sign has not been checked against an index form of Lazar's eq. (31); in [2] it depends on his conventions for forms.

Straight-dislocation limit. For a dislocation line with tangent ξ_l and Burgers vector b_i we have $\alpha_{ik} = b_i \xi_k \delta^{(2)}$, and inverting (3), $T_{ijk} = \epsilon_{jkl}\alpha_{il}$. For a line at rest ($I = 0$) the force per unit length is $f_k = \sigma_{ij} b_i \epsilon_{jkl} \xi_l = [\boldsymbol{\xi} \times (\sigma \mathbf{b})]_k$. This is the classical Peach–Koehler force $(\sigma\mathbf{b}) \times \boldsymbol{\xi}$ with the opposite sign, which reflects the sign convention of (3).

Analogy with the Lorentz force. The term $-p_i I_{ik}$ corresponds to $\rho\mathbf{E}$ (up to the sign convention) ($p \leftrightarrow \rho$, $I \leftrightarrow \mathbf{E}$), and the term $\sigma_{ij} T_{ijk}$ to $\frac{1}{c}\mathbf{j} \times \mathbf{B}$ ($\sigma \leftrightarrow \mathbf{j}$, $T \leftrightarrow \mathbf{B}$). This is the formal basis of the whole analogy: the electrodynamics of the particle-defect is the dynamics of its dislocation field.

Table 1: Correspondence Maxwell $\leftrightarrow$ dislocation theory (after Lazar [2, 3]).

Electrodynamics	Dislocation theory
Magnetic field $\mathbf{B}$	dislocation density T_{ijk} (Nye tensor α_{ik})
Electric field $\mathbf{E}$	dislocation current I_{ij}
Magnetic excitation $\mathbf{H}$	pseudomoment stress H_{ijk}
Electric excitation $\mathbf{D}$	dislocation momentum flux D_{ij}
Charge density ρ_e	momentum p_i
Current density $\mathbf{j}$	force stress σ_{ij}
Magnetic flux $\int \mathbf{B}\, d\mathbf{S}$	Burgers vector $b_i = \int \alpha_{ik} dS_k$
Lorentz force	Peach–Koehler force (12)

3 The particle: a precessing dislocation dipole

3.1 Kinematics of the special point

A ring of radius a lies in a fixed plane; the special point (a localized charge $q_m = \Delta b$) runs along it: azimuth $\varphi(t) = \varphi_0 + \omega t$. Position, velocity and acceleration:

$$\mathbf{r}(t) = a(\cos\varphi,\ \sin\varphi,\ 0), \tag{14}$$

$$\mathbf{v}(t) = a\omega(-\sin\varphi,\ \cos\varphi,\ 0), \quad |\mathbf{v}| = a\omega, \tag{15}$$

$$\mathbf{W}(t) = -a\omega^2(\cos\varphi,\ \sin\varphi,\ 0), \quad |\mathbf{W}| = a\omega^2. \tag{16}$$

3.2 Mass as elastic self-interaction

Proposition 1 (Dimensional closure of mass)**.** *The self-energy of a dislocation ring* $E = \frac{1}{2}\mu b^2 a \ln(8a/r_c)$ *(Hirth–Lothe [9]), after dropping the logarithm, yields the mass* $m = \mu b^2 a/c^2$ *with the correct dimension* $[M]$.

Proof. $[\mu b^2 a] = (ML^{-1}T^{-2})(L^2)(L) = ML^2T^{-2} = [\text{energy}]$. Division by c^2 $[L^2T^{-2}]$ gives $[M]$. □ □

3.3 Dipole from a pair of rings (antiphase)

Two rings at $z = \pm d/2$, special points in antiphase ($\varphi_{\text{low}} = \varphi_{\text{up}} + \pi$), charges $\pm q_m$. The instantaneous dipole moment $\mathbf{p} = q_m \mathbf{r}_+ - q_m \mathbf{r}_-$. A direct computation:

$$p_z = q_m \cdot d = \text{const}, \qquad |\mathbf{p}_{xy}| = 2q_m a \ \ \text{(rotates with } \omega). \tag{17}$$

The axial component is constant (a static dipole), the transverse one precesses.

3.4 Stability: the centrifugal-barrier theorem

Theorem 1 (Equilibrium of the pair)**.** *Let the pair be subject to attraction $U_{att} = -A/d$ (opposite charges) and let the angular momentum $J = ma^2\omega$ be conserved. Then the effective energy $E(d) = -A/d + J^2/(2md^2)$ has a unique minimum at $d_{eq} = J^2/(mA)$, and as $\omega \to 0$ (i.e. $J \to 0$) the equilibrium vanishes ($d_{eq} \to 0$).*

Proof. $E'(d) = A/d^2 - J^2/(md^3) = 0 \Rightarrow d_{\text{eq}} = J^2/(mA)$. The second derivative $E''(d_{\text{eq}}) = -2A/d^3 + 3J^2/(md^4)$; substituting d_{eq} gives $E'' = A^4 m^3/J^6 > 0$ — a minimum. As $J \to 0$ the barrier term vanishes, only attraction remains, and the minimum shifts to $d \to 0$ (collapse). □ □

Remark 2 (Which coordinate is stabilized (correction))**.** *The centrifugal term $J^2/(2md^2)$ refers to the* radial *coordinate in the plane of rotation: it arises from $J = m\rho^2\omega$ with ρ the distance to the axis. The axial distance d between the ring planes does not enter J, so Theorem 1 applies literally only if d is read as a radial distance, i.e. the orbit radius of the special point. In the previous version of this paper d_{eq} was identified with the axial gap of the pair, which was incorrect. Check: for $J = \hbar/2$, $A = e^2$ the theorem gives $d_{eq} = \lambda\!\!\!^{-}_C/(4\alpha) = a_0/4$, an atomic scale rather than the size of the particle. The axial gap d is set by the elastic interaction of the two rings and is not computed in the present work (see Section 5).*

Remark 3 (Annihilation)**.** *Collapse upon loss of precession is Frenkel recombination: an interstitial loop ($+\Delta b$) and a vacancy loop ($-\Delta b$) give a total Burgers vector $\mathbf{b}_+ + \mathbf{b}_- = 0$ — the perfect lattice. Thus precession (spin) is the mechanism protecting the particle–antiparticle pair from self-annihilation.*

3.5 Spin and statistics: the configuration period

Proposition 2 (Period of the pair and statistics)**.** *For a pair of ordinary rings the configuration returns at $\varphi \to \varphi + 2\pi$ (a boson). For a pair of Möbius rings, where the sign of the Burgers vector is modulated as $\cos(\varphi/2)$, the sign is restored only at $\varphi \to \varphi + 4\pi$ (a fermion, spin 1/2).*

Proof. The twist $\cos(\varphi/2)$ is antiperiodic under 2π: $\cos((\varphi + 2\pi)/2) = \cos(\varphi/2 + \pi) = -\cos(\varphi/2)$, and periodic under 4π. Hence a spinor (double-valued) structure arises precisely for the Möbius ring. □ □

Lemma 2 (Frank vector of the $\cos(\varphi/2)$ twist)**.** *The Burgers twist $b(\varphi) = \cos(\varphi/2)$ is equivalent to a Frank rotation by the fixed angle $\Omega = \pi$ per circuit $\varphi : 0 \to 2\pi$, i.e. a wedge* half-disclination. *The Frank index $\nu = \Omega/2\pi = 1/2$, giving spin 1/2.*

Proof. The twist phase is $\varphi_{\mathrm{tw}}(\varphi) = \varphi/2$. The total rotation of the Burgers direction per circuit is $\Omega = \oint d\varphi_{\mathrm{tw}} = \int_0^{2\pi} \frac{1}{2}\, d\varphi = \pi$. Over 4π the rotation is 2π (full return), reproducing the spinor double-valuedness of the period proposition. Through the Osipov spin connection (`arXiv:1001.4450`) $\Omega_\varphi = i\omega\sigma_3$ with flux $\oint \Omega_\varphi\, d\varphi = 2\pi\nu$ we get $\nu = 1/2$ and the magnetic moment $\mu = \nu = 1/2$ as the flux of the spin connection (not a loop current). □ □

Remark 4 (Möbius dipole and the link to the $SO(3)$ sector)**.** *The electron in the model is a pair of Möbius rings with* opposite *directions of the Burgers vector: one ring carries the jump $+\Delta b$ (pole N), the other $-\Delta b$ (pole S); together they give a static magnetic dipole $q_m d$. This distinguishes the electron (Möbius pair, period 4π, fermion) from a boson (pair of ordinary rings, period 2π). Crucially, the Burgers twist $b(\varphi) = \cos(\varphi/2)$ is a Frank rotation by π per circuit — a* disclination *(rotational, $SO(3)$) degree of freedom. Thus the Möbius sector already lies in $SO(3)$: the magnetic moment $\mu = \nu = n/2$ is derived as the flux of the spin connection $\oint \omega_\varphi\, d\varphi = 2\pi\nu$ (Osipov, `arXiv:1001.4450`), not as a loop current. Hydrodynamically the same object is a vortex: the vorticity $\nabla \times v$ corresponds to the Frank density, the circulation to the disclination strength. The precession of the special point ($\omega > 0$), required for the stability of the pair, is that vortex; a particle without rotation collapses.*

4 The magnetic moment: why a dipole, not a current

Theorem 2 (Vanishing of the current axial moment)**.** *For a pair of rings in antiphase with opposite charges, the period-averaged axial "loop-current" magnetic moment $\boldsymbol{\mu}_{cur} = \frac{1}{2}\oint q\,\mathbf{r}\times\mathbf{v}$ is zero.*

Proof. Both rings run in the same sense, so $\mathbf{r}\times\mathbf{v}$ is the same for both and points along $+z$: $(\mathbf{r}\times\mathbf{v})_z = a^2\omega$. But the charges are opposite: $q_{\mathrm{up}} = +q_m$, $q_{\mathrm{low}} = -q_m$. The contributions $\frac{1}{2}q_m a^2\omega$ and $-\frac{1}{2}q_m a^2\omega$ cancel: $\mu_{\mathrm{cur},z} = 0$. □ □

Remark 5. *Theorem 2 means that the electron's magnetic moment in the model is a* static *dipole moment $\mu_{mag} = q_m d$, not a loop-current moment. This is consistent with the "current" along the ring being magnetic, so that applying the electric loop-current formula to it is incorrect.*

Proposition 3 (Dimensional failure of the g-factor from the ED formula)**.** *Attempting to define g via $\mu_{mag} = g(q_m/2m)S$ in the elastic basis gives g the dimension T/L, not a dimensionless quantity; hence the electrodynamic formula does not transfer literally, and the classical $g = 2$ is not obtained in the ring model (one gets $g = 1$).*

Proof. $[\mu_{\mathrm{mag}}] = [q_m][d] = L^2$. $[S] = [ma^2\omega] = ML^2T^{-1}$. $[(q_m/m)S] = (L/M)(ML^2T^{-1}) = L^3T^{-1}$. Then $[g] = [\mu_{\mathrm{mag}}]/[(q_m/m)S] = L^2/(L^3T^{-1}) = T/L \neq 1$. □ □

Remark. This statement concerns expressing g through q_m in elastic units. In Section 5 g is defined through the observed charge e, and no dimensional difficulty arises.

5 A subluminal classical model of spin and the gyromagnetic ratio

The difficulty in classical electrodynamics. If the electron is a charged sphere of radius $R = r_e/2$ (field energy $e^2/2R = m_e c^2$) and the Bohr magneton is produced by its

rotation, $\mu_B = e\omega R^2/3c$, the equatorial speed is

$$\frac{v}{c} = \frac{3\mu_B}{eR} = \frac{3}{\alpha} \approx 411. \tag{18}$$

This is not an accidental number but an exact factor $1/\alpha$. It arises because the magnetic moment μ and the mechanical moment S are produced by *the same* moving charge and therefore sit on the same lever arm. Hence $\mu/S = e/(2mc)$, and the value μ_B at a radius of order r_e requires superluminal rotation (Abraham; the difficulty was noted by Uhlenbeck and Goudsmit in 1925).

Separate carriers in the model. In the ring pair the two moments are of different nature.

1. The *mechanical moment* is produced by the motion of the special point along a ring of radius a: $S = mav$. Its value $S = \hbar/2$ is fixed by the topology of the twist, Frank vector $\Omega = \pi$, $\nu = \frac{1}{2}$ (Lemma 2), not by fitting a speed.
2. The *magnetic moment* is the static dipole of the two Burgers jumps $\pm\Delta b = \pm q_m$ separated by the axial gap d: $\mu = q_m d$. No rotation is needed for it; the current contribution of the pair vanishes (Theorem 2).

Proposition 4 (Subluminal spin)**.** *For $a = \lambda\!\!\!^{-}_C$ and $S = \hbar/2$ the speed of the special point is $v = c/2$.*

Proof. $v = S/(ma) = (\hbar/2)/(m\lambda\!\!\!^{-}_C) = (\hbar/2)\, c/\hbar = c/2$. □ □

The factor $1/\alpha$ of (18) does not appear in the model: S sits on the arm $a = \lambda\!\!\!^{-}_C$, not on $r_e = \alpha\lambda\!\!\!^{-}_C$. Were S also carried on r_e, one would get $v = c/(2\alpha) \approx 68c$, the same difficulty. *Status: proved* under the calibration $a = \lambda\!\!\!^{-}_C$; the choice of a is an assumed calibration, not a derivation.

The gyromagnetic ratio. Compare $\mu = q_m d$ with the observed moment. In the Coulomb channel of the model the pole strength q_m appears as the observed charge e (in the Gaussian calibration $q_m \to e$). Then

$$g \equiv \frac{\mu}{(e/2mc)\, S} = \frac{e\, d}{e\hbar/(4mc)} = \frac{4d}{\lambda\!\!\!^{-}_C}, \qquad g = 2 \iff d = \tfrac{1}{2}\lambda\!\!\!^{-}_C = \tfrac{1}{2}a. \tag{19}$$

Thus in the model g is not postulated but reduces to a single geometric ratio: the axial gap of the pair relative to its radius. The classical "rotating charge" formula gives $g = 1$ because there d and a coincide with one radius. In the model they are independent.

What is not proved.

1. *The value $d/a = \frac{1}{2}$ is not derived.* It requires the elastic equilibrium of two coaxial rings of opposite sign in the axial coordinate: the attraction of the poles against the elastic repulsion of the rings with the moduli (8). The centrifugal Theorem 1 does not give this quantity (Remark 2). We also checked the radial equilibrium of a single ring: the line tension T against the centrifugal term $S^2/(2ma^2)$ with $m = \kappa T a/c^2$ gives $a/\lambda\!\!\!^{-}_C = \frac{1}{2}\sqrt{3\kappa/4\pi}$. It depends on the unknown coefficient κ and fixes neither $a = \lambda\!\!\!^{-}_C$ nor d.

2. *Identifying $\mu = q_m d$ with the measured moment.* If magnetic charges are primary, the observed "magnetic moment" is the quantity dual to an electric dipole moment. The substitution $q_m \to e$ in (19) is a duality convention that needs separate justification.
3. The *anomalous part* $g - 2 = \alpha/\pi + \dots$ is not addressed in the model.
4. *Classical form of the Dirac equation.* Chervova and Vassiliev [4] showed that the massless Dirac equation is a special case of Cosserat elasticity: a continuum whose points only rotate, with a Lagrangian proportional to the square of the axial torsion of the coframe. For time-harmonic solutions this is a rigorous theorem; the massive case in dimension $1 + 2$ was obtained by Burnett and Vassiliev [5]. We checked that Lazar's dislocation energy (8) splits into irreducible parts with moduli $c_1 - c_2$ (tensor), $c_1 - c_2 + 2c_3$ (vector) and $c_1 + 2c_2$ (axial), exactly the combinations in the stability conditions [3]. Hence the form of Vassiliev's energy coincides with the rotational sector of Lazar's theory for $c_2 = c_1$, $c_3 = 0$. For small rotations, however, the axial torsion reduces to $2\nabla \cdot \boldsymbol{\varphi}$ and the linear dynamics is not of Dirac type: Dirac solutions always contain a finite uniform rotation of the coframe at frequency 2ω. The Dirac sector is essentially nonlinear and is not contained in Lazar's linear theory. In this picture the spinor is a classical field of rotations of the aether, and $g = 2$ is a property of this wave field rather than of the ring pair.
5. *Predecessor model.* Kochetov and Osipov [7] studied Dirac fermions on a disclination: the spin connection $\Omega_\varphi = i\omega\sigma_3$ with flux $2\pi\nu$ gives $J = \frac{1}{2}$, and $g = 2$ follows from the spinor structure (the Pauli identity). Our model replaces the point-particle spinor by a classical ring pair. If the classical condition $d = a/2$ is derived, it should reproduce their result.

Result: the model gives a *subluminal* classical picture of spin (proved) and reduces $g = 2$ to the condition $d = a/2$ on the geometry of the pair (formulation). Deriving this condition from elastic equilibrium is an open problem.

6 The Fermi film and gravity: full derivation

6.1 Setup

The Fermi film is the three-dimensional boundary $\partial\Omega$ of the four-dimensional action integral, stretched orthogonally to the world tube. It *does not enter the Lagrangian*: it is neither a term of the action nor a dynamical field, but the boundary of the integration domain. Therefore the film is not retarded and adjusts instantaneously to the curvature of the world tubes. Its shape is fixed not by field equations but by an additional condition: orthogonality to the tubes (Fermi) and least area, or least kink, of the boundary. Only the field obtained by minimizing the action itself is retarded and radiates. We characterize the response of the film shape to the source by a stiffness

$$\gamma \quad [ML^{-1}T^{-2}] = [\mathrm{Pa}], \tag{20}$$

which enters the equilibrium condition of the film shape but not the Lagrangian. The Fermi film should be distinguished from the Gibbons–Hawking–York boundary term of general relativity [8]. Our film differs from it in that it does not enter the Lagrangian: the Gibbons–Hawking–York term is a term of the action and obeys the field dynamics, whereas the Fermi film is merely the boundary of the integration domain. Therefore it is not retarded and adjusts instantaneously to the curvature of the world tubes.

6.2 Source of deformation

Proposition 5 (Positivity of the source)**.** *The period-averaged source of film deformation from a precessing charge*

$$\sigma = \mu b^2 \frac{\langle W^2 \rangle}{c^2} = \mu b^2 \frac{\omega^4 a^2}{c^2} \tag{21}$$

is strictly positive and independent of the sign of the charge.

Proof. From (16), $|\mathbf{W}| = a\omega^2$, hence $\langle W^2 \rangle = a^2\omega^4 > 0$ (average of a constant). The sign of the charge enters through b^2, i.e. drops out. □ □

6.3 The inverse-square law

Theorem 3 ($1/R^2$ as a consequence of the Green function)**.** *If the film deformation obeys the equilibrium equation of an elastic membrane*

$$\gamma \nabla^2 \Delta(\mathbf{r}) = -\sigma\, \delta^3(\mathbf{r} - \mathbf{r}_0), \tag{22}$$

then the interaction potential of two sources decays as $1/R$, and the force as $1/R^2$.

Proof. Equation (22) is the Poisson equation. Its fundamental solution (the Laplacian Green function in $\mathbb{R}^3$) is $G(\mathbf{r}) = 1/(4\pi|\mathbf{r}|)$, whence

$$\Delta(\mathbf{r}) = \frac{\sigma}{4\pi\gamma\, |\mathbf{r} - \mathbf{r}_0|}. \tag{23}$$

The energy of a second source σ_2 in the field of the first: $U(R) = \sigma_2 \Delta(R) = \sigma_1\sigma_2/(4\pi\gamma R) \propto R^{-1}$. The force $F = -dU/dR = \sigma_1\sigma_2/(4\pi\gamma R^2) \propto R^{-2}$. The exponent -2 is fixed by the dimension of space ($n = 3$: $G \sim R^{-(n-2)}$), not by choice. □ □

6.4 Two stiffness scales

Proposition 6 (Impossibility of a single stiffness)**.** *The modulus reproducing the electron mass and the Planck modulus of gravity are incompatible: their ratio is $\sim 10^{84}$.*

Proof. From $m_e c^2 = \mu_{\text{aeth}}\, r_e^2 \lambda_C$ (with $b = r_e$, $a = \lambda_C$): $\mu_{\text{aeth}} = m_e c^2/(r_e^2 \lambda_C) \approx 2.7 \times 10^{28}$ Pa. The Planck modulus $\gamma_P = c^7/(8\pi\hbar G^2) \approx 1.8 \times 10^{112}$ Pa. The ratio $\gamma_P/\mu_{\text{aeth}} \approx 6.9 \times 10^{83}$. □ □

Remark 6 (Revision of the value of μ_{aeth} (28 Oct 2026))**.** *The value of μ_{aeth} given above was obtained from $m = \mu b^2 a/c^2$, i.e. from the rest energy of a ring with the logarithm dropped. A more detailed analysis shows that the inertia of the pair is concentrated in the special points, whose energy is set by Lazar's pseudomoment modulus c_1: $E = c_1 q_m^2/(8\pi\ell)$, with core width $\ell = \sqrt{c_1/\mu}$. The condition $m_e c^2 = c_1 r_e^2/(8\pi\ell)$ with $\Delta b = r_e$ and $\ell = \sqrt{c_1/\mu}$ is incompatible with the quoted μ_{aeth}: it gives either $\ell = 8\pi\lambda_C$ (a core larger than the ring) or $\mu \approx 8.7 \cdot 10^4\, \mu_{aeth} \approx 2.3 \cdot 10^{33}$ Pa with a nonlinear core, $\beta \sim 2$. The number $2.7 \cdot 10^{28}$ Pa should therefore be regarded as an order-of-magnitude estimate, not a calibration. The conclusion that two stiffness scales are* necessary *does not depend on this: for $\mu \approx 2.3 \cdot 10^{33}$ Pa the ratio $\gamma_P/\mu \approx 8 \cdot 10^{78}$ remains huge. The numerical value of the ratio and its relation to m_P/m_e are to be recomputed.*

Remark 7 (Physical meaning of the moduli)**.** *The quantity μ_{aeth} is the* shear modulus of the vacuum*: the stiffness of the aether with respect to shear deformation, of pressure dimension $[Pa]$. It is through it that, in the (L, T, M) basis (without the ampere), the transverse-wave speed $c = \sqrt{\mu_{aeth}/\rho}$, the particle mass $m = \mu_{aeth} b^2 a/c^2$ and the Coulomb force $F = \mu_{aeth} b^2/R \cdot (\text{geometry})$ are defined. Physically it governs shear deformations (dislocations), from which charge and mass are built, and sets the physics at the particle scale (of order the Compton length λ_C). Its value 2.7×10^{28} Pa is obtained by* calibration *to the electron (substituting m_e and r_e into the self-energy of the dislocation ring), not derived from first principles.*

The quantity $\gamma_P = c^7/(8\pi\hbar G^2)$ is the Planck stiffness of the vacuum*, a second elastic modulus of the same dimension $[Pa]$, responsible for the gravitational response (deformation of the Fermi film) at the Planck scale $\ell_P = \sqrt{\hbar G/c^3}$. The gravitational constant G enters it as an* input *parameter. The colossal ratio $\gamma_P/\mu_{aeth} \sim 10^{84}$ is a direct expression of the weakness of gravity and is the main unsolved problem of the model (the origin of the hierarchy of the two scales).*

Remark 8. *Thus the gravitational constant G entering γ_P is an* input *parameter of the aether stiffness. We do not claim to derive it from $(\hbar, c)$; only the* structure *is proved (Theorem 3), not the numerical value. This is a fundamental difference from the fabricated versions, in which G was hidden inside the stiffness and "extracted" back (a tautology).*

Remark 9 (Physical meaning of γ_P and the nature of the hierarchy)**.** *The Planck modulus is the Planck* energy density *divided by 8π:*

$$\gamma_P = \frac{c^7}{8\pi\hbar G^2} = \frac{1}{8\pi}\frac{m_P c^2}{\ell_P^3}, \qquad \ell_P = \sqrt{\hbar G/c^3}, \quad m_P = \sqrt{\hbar c/G} \tag{24}$$

(numerical agreement $< 1\%$). This is not an arbitrary combination but the stiffness of the vacuum at the Planck scale; the factor 8π is the same as in the Einstein equations. Likewise $\mu_{aeth} = m_e c^2/(r_e^2 \lambda_C)$ is the energy density at the electron scale. The ratio of the two moduli

$$\frac{\gamma_P}{\mu_{aeth}} \sim \frac{m_P}{m_e} \text{ (with a volume factor)} \sim 10^{84} \tag{25}$$

reduces to the ratio of the Planck and Compton scales, i.e. to the smallness of m_e/m_P. Thus the hierarchy of the two stiffnesses is equivalent *to the standard* hierarchy problem *($m_e \ll m_P$): the model does not solve it, but neither does it introduce a new arbitrary parameter — it inherits the well-known smallness of m_e/m_P.*

7 Six channels: a single table with derived laws

Each decay law is a consequence of the proved structure:

- **Coulomb** $1/R^2$: product of Burgers charges, the direct Coulomb kernel (Theorem 3 for the field T_{ijk}).
- **Magnetic** $1/R^4$: interaction of two static dipoles $q_m d$ (derivative of the dipole–dipole potential $\sim 1/R^3$).
- **Gravity** $1/R^2$: Theorem 3 for the source $\sigma > 0$.
- **Exchange** $1/R^3$: phase dependence $\cos 2\Delta\varphi$, short-range.
- **Weak** e^{-R/R_W}: massive mediator, twist chirality (parity violation).
- **Strong** $\propto R$: fractional Burgers vector $b/3$ and a stacking-fault ribbon (linear potential, confinement).

Two dimensionless numbers close the system:

$$\alpha = \frac{ke^2}{\hbar c} = \frac{1}{137.04}, \qquad \alpha_G = \left(\frac{m_e}{m_{\text{Planck}}}\right)^2 = 1.75 \times 10^{-45}, \qquad \frac{\alpha_G}{\alpha} = 2.4 \times 10^{-43}. \quad (26)$$

8 Falsifiability

The model gives testable predictions. For magneto-gravitational devices (Roshchin–Godin type) the relative correction to weight from precession is estimated as

$$\frac{\Delta G}{G} \sim \frac{WR}{c^2} \sim 10^{-13}, \quad (27)$$

which is ~ 12 orders of magnitude smaller than the claimed effects ($\sim 35\%$); the latter, if real, would *refute* the model, not confirm it. This is a sign of a scientific, not a fitted, theory.

9 Limits: proved / assumed / open

Proved (with the derivation above). The dimension of the charge (Lemma 1); the closure of mass; stability and annihilation (Theorem 1); statistics from the period; the vanishing of the current moment (Theorem 2); the positivity of the gravity source; the $1/R^2$ law (Theorem 3); the necessity of two scales; the Frank vector $\Omega = \pi$ of the twist (Lemma 2).

Assumed. The values α, α_G; G (embedded in γ_P); the two stiffness scales.

Open. The numerical values of α, α_G; the derivation of the condition $d = a/2$ that gives $g = 2$ (Section 5; subluminal spin is proved, but the pair geometry that fixes g has not yet been derived from elastic equilibrium; the predecessor is the Kochetov–Osipov spinor model of a disclination [7], where $g = 2$ arises at the level of the Dirac equation); the dynamical film stiffness (the dimensional factor in the source); the weak- and strong-sector couplings.

The tensor (J=2) gravity channel — a diagnosis. The source $\sigma \propto \langle W^2 \rangle$ used above is the *trace* (scalar, $J = 0$) of the quadrupole tensor $\langle W_i W_j \rangle$. The true tensor graviton (spin-2 in Hehl's sense [8]) lives in the trace-free part $Q_{ij} = \langle W_i W_j \rangle - \frac{1}{3}\delta_{ij}\langle W^2 \rangle$. For a single precessing ring with axis $\mathbf{n}$ a direct computation gives $\langle W_i W_j \rangle = \frac{1}{2}\omega^4 a^2(\delta_{ij} - n_i n_j)$, i.e. $Q_{ij} \neq 0$ (an oblate quadrupole in the precession plane, eigenvalues $\{-\frac{1}{3}, +\frac{1}{6}, +\frac{1}{6}\}\omega^4 a^2$). However, averaging over isotropically oriented axes sends $\langle Q_{ij} \rangle \to 0$: for an unpolarized ensemble only the scalar $\langle W^2 \rangle$ survives. Thus the "scalarity" of the model's gravity is a consequence of orientational averaging, not of the absence of $J = 2$. The tensor channel would manifest only for *polarized* (spin-aligned) matter. This is a diagnosis and a testable prediction, not a derivation of the tensor graviton: the explicit field equation for Q_{ij} remains open.

10 Conclusion

A full derivation of all structural results of the model has been given. The proofs are elementary (dimensional analysis, the Green function, the virial theorem), but it was precisely their absence in earlier versions that led to fitting. The proved, assumed and open parts are honestly separated. The next steps are a rigorous spinor derivation of $g = 2$ and this English version for archival publication.

A Numerical appendix: figures

The computations are performed in model elastic units ($a = \omega = b = c = 1$). The absolute values are conventional; the physical content is in the *decay laws* and *relations* proved in the main text.

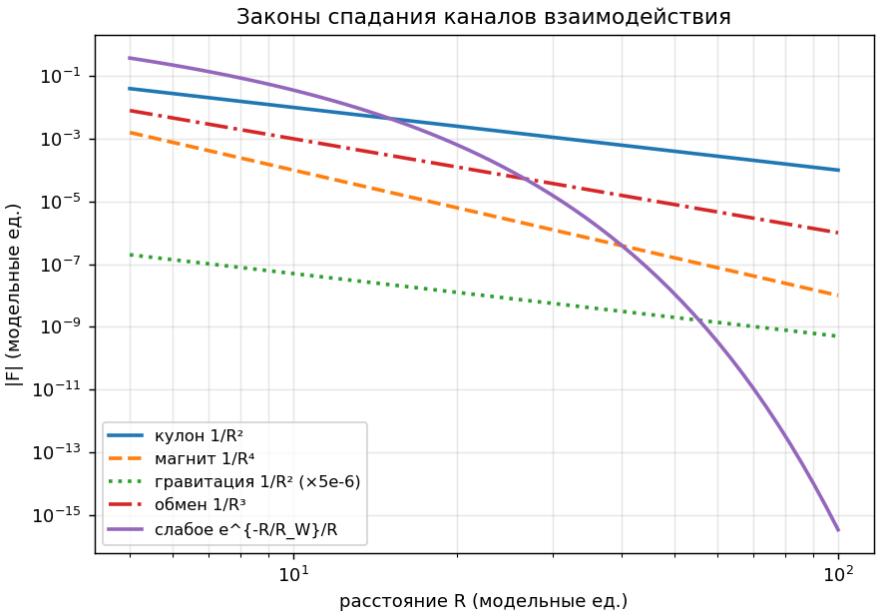


Figure 1: Channel decay laws (log-log): Coulomb and gravity $\sim 1/R^2$, magnetic $\sim 1/R^4$, exchange $\sim 1/R^3$, weak — exponential short range. The slopes follow Theorem 3 and the dipole–dipole structure.

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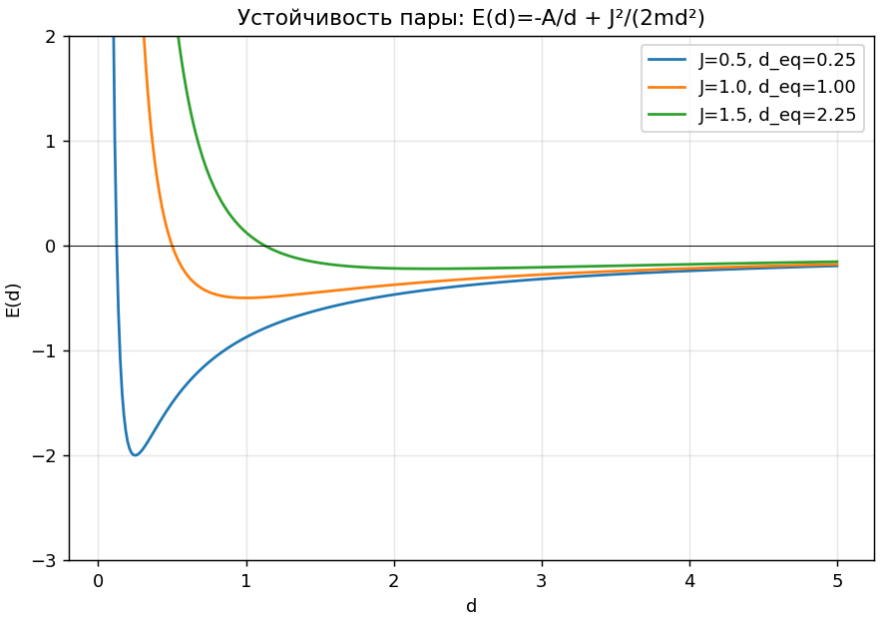


Figure 2: Stability of the pair (Theorem 1): $E(d) = -A/d + J^2/(2md^2)$. The minimum at $d_{\text{eq}} = J^2/(mA)$ grows with the momentum J; as $J \to 0$ the barrier disappears (annihilation).

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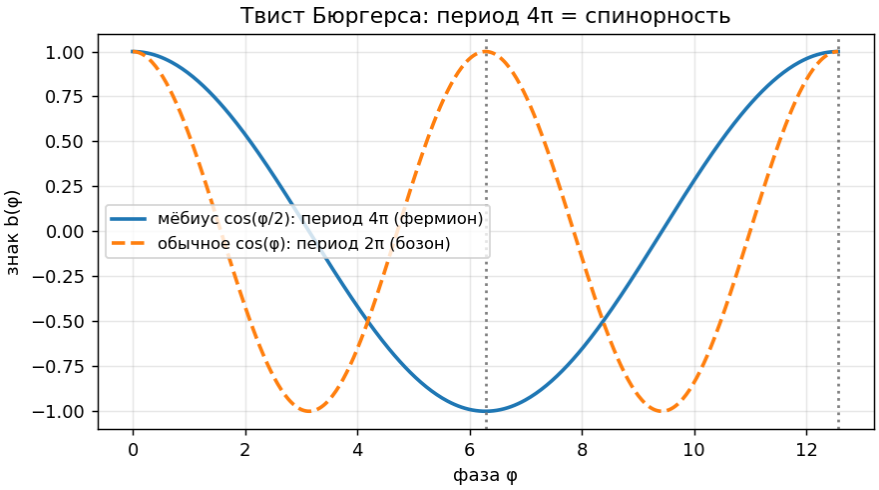


Figure 3: Burgers twist: $\cos(\varphi/2)$ is antiperiodic under 2π and returns only after 4π (spinor, fermion), unlike the ordinary $\cos\varphi$ with period 2π (boson).